

EDUCATION

University of Miami - MFA in Interactive Media

Aug 2018 – Dec 2020

GPA: 3.50 / 4.0

Miami, United States

Key Courses: Geographic Information Systems (A), 3D Design (A), Dynamic Data (A+), Mobile App Development (A)

Capstone: Data visualization of *The Twelve Kingdoms*. Created a repeatable pipeline that connected episode-level "bullet screens" telemetry and novel text analysis; produced an interactive visualization that showed narrative beats and engagement spikes. [[Project](#)] [[Code](#)] [[Process](#)]

China University of Political Science and Law - Bachelor of Sciences

Sep 2013 – July 2017

GPA: 3.34 / 4.0

Beijing, China

RESEARCH EXPERIENCES

3D Interactive Origin-Destination Dashboard [[Demo](#)] [[Code](#)] [[Process](#)]

Nov 2025

- Participated in five product and UI meetings as the data visualization engineer to define OD encodings, leading to the adoption of 3D pillars and arclines.
- Led front-end research on 3D visualization options and selected a MapLibre.js + Deck.gl pipeline for interactive OD rendering.
- Designed and implemented a separate 2D engine that renders millions of heatmap data points with 50-millisecond refresh for real-time monitoring.
- Received strong recognition from leadership and clients for OD insight, visual clarity, and interaction performance.

Intersection AI Agent Interface [[Demo Video](#)] [[Process](#)]

Sep 2025

- As lead UI/UX designer and front-end engineer for the Hangzhou Intersection AI Agent, I translated a complex traffic-control workflow into a clear five-step interactive loop: sense → propose → compare → deploy → review.
- I designed visual interfaces that expose the agent's internal state, scoring logic, and decision history, improving transparency and auditability for operators.
- I implemented real-time visual overlays, including signal plans, queue lengths, and before/after comparisons, so that operators can immediately verify the effects of traffic signal changes.
- By enabling full automation for seven intersections and replacing continuous CCTV monitoring with intuitive, explainable dashboards, the system significantly reduced manual workload and enhanced operational efficiency.

Real-time Intersection Information Visualization [[Demo Video](#)] [[Process](#)]

Mar 2025

- Led the design and front-end development of a Holographic Intersection Dashboard, creating real-time 2D overlays to display intricate traffic data, including speed, vehicle flow, and signal countdowns.
- Compared to traditional CCTV-only systems, a quasi-experimental evaluation showed how well the system improved operator awareness.

- WebAssembly frame interpolation and layered-canvas optimization were used to achieve stable 60 FPS rendering.
- Reduced operator task time and workload (TLX), shorter wait times, and 17% faster vehicle approach speeds were noted.

Auto-Visualization for Geospatial Workflows [[Demo Video](#)] [[Process](#)]

Feb 2025

- Led the design and front-end development of an auto-visualization workspace that processes and displays intricate, city-scale geospatial data automatically.
- Through a thorough A/B test, the system's efficacy was confirmed, demonstrating its capacity to improve data accessibility and produce instantaneous, executive-ready views.
- provided streaming loaders and declarative, type-driven mappers (converting varchar to categorical and integer to threshold, gradient, or heatmap).
- TTFI and task time dropped, strategic decision-making was expedited, large-scale rendering was optimized, and reporting was completed 35% faster.

Point-to-Video: Automated Vehicle Trajectory Pipeline [[Code](#)]

Apr 2024

- Be responsible for the design, development, and automation of a Point-to-Video vehicle-trajectory pipeline that transformed time-stamped points into repeatable animations.
- provided region-aware color mapping, headless per-slice capture with image assembly, and deterministic fixed-timestep interpolation with ID stitching.

Gamified Green-Wave Optimization Exhibit [[Project](#)] [[Process](#)]

Feb 2024

- Created and implemented a gamified, interactive layer to replace static dashboards at client conferences and headquarters.
- Enhanced understanding of stop-rate and green-wave, increasing customer interaction and collaboration prospects.

What Winter Olympic Sports Are You Suitable For? [[Project](#)] [[Process](#)]

Feb 2022

- Led the end-to-end design and launch of an interactive Beijing 2022 quiz—data collection, question design, UI, and micro-animations.
- Built a responsive front end with a recommendation engine that maps answers to sport profiles and playful result visuals.
- Featured on Caixin Data Visualization Lab; drove high user engagement and share-through, strengthening brand visibility.

Where Did Your Daily Wastewater Go? [[Project](#)]

Apr 2019

- Led the design and production of a 3D infographic explaining municipal wastewater treatment and downstream environmental effects.
- Built a concept-to-render pipeline: research & storyboard, Illustrator iconography, Maya modeling, lighting, and compositing.

WORK EXPERIENCES

Tranalytic Technology Co., Ltd. (Subsidiary of Merit Interactive Co., Ltd.)

Oct 2022 – Current

Front-End & Data Visualization Engineer

Hangzhou, China

Yutong Han

<https://yutonghan19951123.github.io/> • hanyutong19951123@gmail.com

- Led the development of real-time, smart-traffic visualization platforms, reducing dashboard latency by 40% and increasing decision-making efficiency by 35% for over 50 analysts and executives.
- Collaborated with product and algorithm teams to build AI agents and City Brain-style systems, slashing data-to-visualization turnaround from months to a single month to power high-impact client showcases.
- Architected a reusable front-end architecture and component library, reducing redundant development work by 35% and enabling scalable, multi-city deployments.

Caixin Media Company Ltd.

Mar 2021 – Sep 2022

Data Journalist

Beijing, China

- Directed the end-to-end lifecycle of over 100 data stories, from initial data collection to final publication, by analyzing large-scale, multi-source datasets and relevant policy contexts.
- Designed interactive visualizations and infographics for news narratives, collaborating with designers to deliver locally focused explainers that averaged over 10,000 readers per feature.

SPEAKING

How to Use WebGIS Gracefully [[PPT](#)]

June 2024

Geospatial Data Visualization—From Design to Implementation [[PPT](#)]

June 2024

The Design System of Tranalytic Technology

Feb 2023

What are heat maps, tile diagrams and Sankey diagrams

Nov 2022

JOURNALISTIC & PUBLIC-FACING PUBLICATIONS

Is the NFT boom a flash in the pan [[Report](#)]

Caixin, Sep 7, 2022

Why did heat waves persist? [[Report](#)]

Caixin, Jul 22, 2022

Beijing Universal's Bind: What hurts beyond COVID? [[Report](#)]

Caixin, Jun 28, 2022

What Makes Ice Sports Popular Worldwide [[Report](#)]

Caixin, Feb 1, 2022

“Youth With You 3” Halted: Inside China’s Fandom Data-Work [[Report](#)]

Caixin, Apr 25, 2021

“Hi, Mom” crosses ¥50.98B: the women behind the big screen [[Report](#)]

Caixin, Mar 8, 2021

The “secret” to fund gains [[Report](#)] [[Game](#)]

Caixin, Mar 3, 2021

SKILLS

Front-End Development: Proficient in HTML5, CSS3, JavaScript (ES6+), TypeScript, and frameworks including Vue 3 (Composition API, Vuex, Nuxt.js), React (Hooks, Redux), and Next.js; skilled in responsive design with Tailwind CSS, Bootstrap, and Sass/Less.

Geospatial Visualization: Experienced in interactive dashboards using MapLibre, ECharts, AntV, and Canvas, enabling real-time rendering of large-scale data; familiar with Three.js and Canvas for 3D/2D visualization.

UI/UX Design & Prototyping: Skilled in Figma, Sketch, Adobe Illustrator, and Animate for user-centric interfaces and interactive prototypes.

Performance & Architecture: Specialized in high-performance visualization components and front-end optimization; familiar with HTTP protocols and webpack/Vite for scalable builds.

Data Processing: Excel, MySQL, R, and Python

Language: English, Mandarin Chinese

ACHIEVEMENTS

Third Prize (Group)	2024
Yangtze River Delta Intelligent Transportation Innovation Technology Application Competition	
Second Prize (Group)	2024
National ‘Data Elements ×’ Competition – National Finals	
Excellence Award , Tranalytic Technology Co., Ltd.	2024
Recognized for outstanding contributions to data visualization and front-end engineering within the Smart Traffic division.	
Best Newcomer Award , Tranalytic Technology Co., Ltd.	2023
Awarded for exceptional performance and rapid adaptation during the first year at the company.	